



OFFICIAL

Blood Management Manual

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1. Applicability

Site	Operational Area	Applicable to
Armadale Kalamunda Group	All areas	Medical, Nursing, Patient Support Services

2. Purpose

To ensure safe, appropriate, efficient, and effective care of patients own blood, as well as other blood and blood products and ensure the workforce uses the blood product safety systems.

iTransfuse App



The Armadale Kalamunda Group endorses the use of the iTransfuse App developed by the Australian Red Cross. The iTransfuse App promotes safe, appropriate, and evidence-based transfusion by providing bedside access to transfusion education tools and resources.

The app can be downloaded free from the App Store or Google Play

3. Decision to transfuse

Every ONE matters

The decision on whether to transfuse should be carefully considered considering the full range of available therapies and balancing the evidence for efficacy and improved clinical outcome against the potential risks.

Clinicians should always aim to avoid or minimise the use of allogeneic blood components, both in the patient's interests and to avoid wastage of a limited and valuable resource.

[ROTEM see Appendix 1](#)

Transfusion should not be a default decision. It should not be dictated by a Hb concentration alone but should also be based on assessment of the patient's clinical status.

The decision to transfuse must be based on a thorough clinical assessment of the patient and individual needs.

The indication for transfusion, or other blood management strategies chosen, must be discussed with the patient as part of their care plan and documented in the patient's medical notes. Consider alternatives to red cell transfusion such as iron therapy, B12 folate in patients who are deficient.

This applies to all patient populations:

- In the perioperative patient, the program encourages targeting preoperative optimisation of red cell mass and coagulation status, minimising operative blood loss and tolerance of postoperative anaemia [Perioperative Blood Management Guidelines](#)
- In medical patients, consideration of the patient's medical conditions and clinical status is critical in dictating the need for transfusion [Medical Blood Management Guidelines](#)
In major haemorrhage for obstetric or non-obstetric the Senior clinician determines that the patient meets criteria for Massive Transfusion [Protocol Critical Bleeding/Massive Transfusion Blood Management Guidelines](#)
- In critical patients, consideration of the patient's clinical status will guide transfusion requirements rather than the haemoglobin level in isolation [Critical Care Blood Management Guidelines](#)
- In managing pregnant and postpartum women [Obstetrics and Maternity Blood Management Guidelines](#)
- In the paediatric and neonatal patient provides guidance on dosing, assessment and optimisation of various products and situations in the neonatal and paediatric population [Neonatal and Paediatrics Blood Management Guidelines](#)

Hb > 90g/L	Transfusion is likely to be unnecessary and inappropriate unless the patient is actively bleeding . RBC transfusion in Hb>100g/L is associated with increased mortality in patients with acute coronary syndrome (PBM guide: Medical).
Hb 70-90g/L	Medical patients: Transfuse to relieve sign and symptoms of anaemia. Consider 1unit RBC transfusion and reassess patient. No evidence exists for a different approach for elderly patients or those who have respiratory or cerebrovascular disease. Surgical patients: Post op patients with acute MI or cerebrovascular ischaemia: Transfuse 1-unit RBC and reassess.
Hb < 70g/L	RBC Transfusion maybe appropriate to relieve the clinical signs and symptoms of anaemia after excluding other causes for clinical presentation. Transfusion may not be required in well-compensated patients.

4. Fresh blood products and indication for use

Product		Indication for Use
Red Blood Cells (RBCs)		RBCs are extracted from whole blood by centrifuge. They are used to treat clinically significant anaemia where the patient is actively symptomatic or where there is active bleeding from trauma or surgery. RBCs are provided in blood groups O, A, B and AB, and Rh (D) negative or positive. It is critical that the blood group is compatible with the patient's blood group. In emergencies, a crossmatch sample should be taken, and then O-negative blood may be administered.
Platelets		Platelets derived from whole blood and are used to treat bleeding associated with surgery, massive haemorrhage, platelet dysfunction, or for patients with bone marrow failure at high risk of haemorrhage. Must be ordered "as required" for specific cases from Red Cross Blood Transfusion Service.
Fresh Frozen Plasma (FFP)		FFP is derived from whole blood and contains all the clotting factors, including a large volume of Factor VIII and Factor V. It is used to correct coagulopathies such as occur in massive transfusion, cardiac bypass, liver disease or in disseminated intravascular coagulation.
Cryoprecipitate		Derived from the precipitate left when fresh frozen plasma is thawed at a low temperature. It contains Factor VIII, fibrinogen, Factor XIII, von Willebrand's factor and fibronectin. It is used when there is low fibrinogen associated with major haemorrhage, an invasive procedure or when clotting factors are being massively consumed in disseminated intravascular coagulation (DIC).

5. Paediatric and neonatal blood management

Armadale Health Service follow the Transfusion Medicine Protocols developed and endorsed by Perth Children's Hospital (PCH) for use in Clinical Practice for paediatric patients.

For consent, prescriptions and documentation use the available forms at AHS.

[Perth Children's Hospital Transfusion Medicine Protocols](#)

Armadale Health Service follow the policies and guidelines developed and endorsed by King Edward for use in Clinical Practice for Neonatal Patients.

[Women and Newborn Health Service Transfusion Protocols \(Neonates\)](#)

6. Pre-transfusion requests and sample collection

6.1 Key points

- Correctly identify the patient by asking them to state their name and date of birth, and check against their identity band – for unconfirmed identity and name changes, refer to section of “Unconfirmed Identity and Name Changes” in this document.
- Complete Pathology Request, preferably via e-order or CPOE, if unavailable use PathWest Pathology Request form.
- Obtain patient transfusion history. Document on request form if patient pregnant, gestation, recent administration of RhD Immunoglobulin.
- Neonatal Crossmatch requests (<4months of age) requires a valid blood group and antibody screen (GAS) from the Mother and a sample from the baby for Direct Antiglobulin Test).
- Collect, or arrange collection of samples: GAS 6mL EDTA – pink top, and label sample and request with patient identifiers (Family name, given name, DoB, UMRN).
- **Sample: Sign** and write **date and time on sample**.
- **Request: Sign** collector declaration (signature or initials must match sample) at point of collection. (bedside)
- **Request: Write Date and time** at collector declaration.
- **Refer** to general Pathology Request Form and Transfusion medicine request form for PathWest examples of Transfusion Requests.

PathWest LABORATORY MEDICINE WA
Hospital Avenue
Nedlands WA 6009
ABN 83 489 340 804

RESULTS & ENQUIRIES 13 PATH 7284

PATHOLOGY REQUEST

CPOE DOWN-TIME FORM

Unit no. A1212122 Medicare Number _____
Surname PATIENT
Given Names TEST
Date of Birth 2-2-90 Age _____ Sex M
Address _____

TESTS REQUESTED ☒ URGENT ☐ PHONE ☐ FAX ☐
Ph / Fax Number: _____

Group and Hold

The signature/initials in the declaration box must match the signature/initials on the blood tube EXACTLY

Consultant A. Consultant
Requesting Doctor (surname and initials, provider number, address) B. Doctor 2987654P AHS
Request Date 26.6.23 Page 123
Copy Reports to: _____

Doctors Signature X B. Doe

Source / Hospital ARMA Ward / Clinic E.D
Day For Collection M T W Thu F S Su
When collecting ANTIBIOTIC or DRUG assays fill in this box
Drug Dosage Date Time
Date of Collection 26.6.23 Time of Collection 1200
ACD HEP EDTA
GLU ESR ABG
URINE 24 URINE SWAB
SLIDE Other 1 Pink

Collector's Signature
I certify that the blood specimen(s) accompanying this request was drawn from the patient named above and I established the identity of this patient by direct inquiry (the blood being drawn / labelled the specimen)

X B. Doe

specimens collected:
1. A private patient in a private hospital or approved day hospital facility ☐
2. A private patient in a recognised hospital ☐
3. A Medicare (public patient) in a recognised hospital ☐
4. An outpatient of a recognised hospital ☐

Results to be: ☐ Faxed ☐ Phoned

Patient's Signature for Ancillary Test: X

- **Blood orders:** include product, number units/dose, date, time, and location of planned transfusion or procedure, any special requirements, reason for transfusion.
- If **patient identity changes** after collection of pre-transfusion sample, you **must** notify PathWest Laboratory immediately.
- If **collector error** is realised after collection of pre-transfusion sample, you **must** notify PathWest Laboratory immediately.
- Re-labelling, or correcting labels on samples or requests is not permitted. If samples do not meet the correct labelling guidelines as described above the request will be rejected and a new collection of sample and form will be required. This will be strictly adhered to even if the case is extremely urgent.
- A GAS can be completed in 45 minutes if urgent blood is required – specify if critically bleeding.
- Sample validity:
 - 72 hours if transfused in last 3 months
 - 7 days if pregnant or in the immediate post-partum

Note: Failure to correctly identify the patient (e.g. at specimen collection or before transfusion), prescribing the wrong product and transfusing the wrong patient are all causes of major morbidity and mortality.

- Patients are identified according to the [EMHS Patient Identification and Procedure Matching](#)

Samples and Requests for pre-transfusion testing and antenatal blood group and antibody screening are collected, labelled and signed in accordance with the National Pathology Accreditation Advisory Council (NPAAC) *Standard for Transfusion Laboratory Practice* and the Australian and New Zealand Society of Blood Transfusion (ANZSBT) [Guidelines for Transfusion and Immunohaematology Laboratory Practice](#). For sample tube information – refer to the [PathWest Test Catalogue](#)

6.2 Responsibilities

- All clinical staff ordering pre-transfusion, antenatal and cord blood testing must provide the required information and correctly complete the request form for sample collection and blood ordering.
- All staff collecting pre-transfusion, antenatal and cord samples must adhere to the collection and labelling requirements outlined in this policy.
- All laboratory staff registering and processing samples for pre-transfusion, antenatal and cord blood group requests must check the samples and requests match and meet the labelling requirements outlined in this manual.
- Samples and/or requests that do not meet the requirements or that do not match, cannot be processed. The collector/requestor will be notified to organise a re-collection – a new request form will be required.

6.3 Patient Identification

Before commencing collection, ask patient to state:

- Family name
- Given name
- Date of birth

- **AND** check these 3 patient identifiers against the identification band attached to patient to ensure they match
- A unique identifier must be used (UMRN) for sample and request labelling and these details linked to the patient's name

6.4 Process for unconfirmed identity and name changes

When a patient's name cannot be reliably confirmed by the patient or next of kin, e.g. temporary or long term intellectual impairment, inability to communicate or loss of consciousness, they must be registered as per hospital policy as "Unknown Male" or "Unknown Female" using an emergency UMRN number. These details must be used for labelling sample and request.

Once the patient is identified, patient information should be updated in WebPAS/TOPAS and a new identification band attached.

If the identity is confirmed during a critical bleeding situation or critical medical management, e.g., resuscitation or surgery, then the change of identity **should** occur only when the period of potential/actual critical bleeding has ended and no longer requires transfusion support. See [EMHS Patient identification and Procedure Matching Policy](#)

6.5 Cord blood and neonatal testing

The request form and sample must clearly identify the baby and include information linking the baby to its mother. All other processes set out in this policy for collection apply. For example.

- Single births (Singleton)
- Baby UMRN
- Mother's FAMILY NAME, Baby of [mother's given name, mother's UMRN]
- Baby's date of birth and gender

If cord sample is collected into a pink EDTA tube it must be clearly labelled "CORD"

Multiple births information is required in addition to above i.e., for twins: TW1, TW2: for triplets: TR1, TR2, TR3.

6.6 Requests

All requests for pre-transfusion testing must be written and signed by the requesting medical officer. Electronic CPOE requests will insert requesting medical officer electronically and are preferred.

The following information is required:

- Patient family name, full given name, hospital record number, date of birth, gender
- Name, signature, and provider number (electronic if CPOE) of requesting medical officer
- Details of the request
 - Test required (Group and Screen is required for blood transfusion and antenatal testing, Group and Direct Antiglobulin Test (DAT) for Cords and Neonates)
 - Type of blood product requested
 - Number of units/doses for vials - note for pre-Surgical patients and date required
 - Clinical diagnosis and indication with supporting laboratory data
 - Known red cell antibodies

- Obstetric history including date and dose of recent RhD immunoglobulin administration i.e., on current pregnancy

6.7 Sample, labelling and signing

6.7.1 Sample

Group and Screen (GAS), Blood Group, Crossmatch

- 6mL EDTA sample (1 pink top) – refer to [PathWest Test Catalogue](#) for further information and for other test collections.
- Group and Screen sample is valid for 14 days if patient is not pregnant and not transfused in the last 3 months.
- If patient is pregnant the sample is valid for 7 days.
- If patient transfused in last 3 months and/or has a positive antibody history, the sample is valid for 72 hours

6.7.2 Labelling

Before leaving the patient, ensure complete and accurate labelling of sample following collection (addressograph labels or handwritten are acceptable).

The sample must have:

- Patient family name and given name in full
- Medical record number
- Date and time of collection
- Collector's signature (initials/signature to match collector/requestor declaration signature on request form). If initials are used on the declaration, then initials should be used on the sample.

PathWest Hospital Avenue Nedlands WA 6009 ABN 83 489 340 804

RESULTS & ENQUIRIES 13 PATH 7284

PATHOLOGY REQUEST

CPOE DOWN-TIME FORM

Unit no. A12122 Medicare Number _____

Surname PATIENT

Given Names Test

Date of Birth 2-2-90 Age _____ Sex M

Address _____

TESTS REQUESTED _____

URGENT ☒ PHONE ☐ FAX ☐

Ph / Fax Number: _____

Group and Hold

The signature/initials in the declaration box must match the signature/initials on the blood tube EXACTLY

CLINICAL NOTES _____

Results to be: ☐ Faxed ☐ Phoned

Consultant A. Consultant

Requesting Doctor (surname and initials, provider number, address) B. Doctor
Z987654P
A45

Source / Hospital ARMA Ward / Clinic E.D

Day For Collection _____

When collecting ANTIBIOTIC or DRUG assays fill in this box:

Drug	Dosage	Date	Time

Date of Collection 26-6-23 Time of Collection 1200

TEST	SST	CIT
ACD	HEP	EDTA
GLU	ESR	ABG
URINE	24 URINE	SWAB
SLIDE	Other	<u>1 Pink</u>

Collector's Signature / I certify that the blood specimen(s) accompanying this request was drawn from the patient named above and I established the identity of this patient by direct inquiry the blood being drawn / labelled the specimen(s) X B. Doc.

Request Date 26-6-23 Page 123

Copy Reports to: _____

Fasting: Yes ☐ No ☐

Rule 3 Exemption: Yes ☐ No ☐

Anticoagulant Therapy Warfarin ☐ Heparin ☐

Patient's Signature for Ancillary Test: X

Tube Label: A12122
PATIENT Test
26-6-23 1200
B. Doc.

6.7.3 Signing

For all pre-transfusion requests, the sample and the request must be signed by the collector and date/time of collection recorded.

The Transfusion Request and the CPOE request will have the following statement:

I certify that I collected the accompanying sample(s) from the above patient, whose identity I confirmed by direct inquiry and/or examination of their name-band, and that I labelled the sample(s) immediately following collection, and signed the label(s), and wrote the date and time of collection on the label(s).

Signature..... Name.....Date collected.....Time collected.....

6.7.4 Verbal requests for blood

Verbal requests for blood for an urgent blood/blood products transfusion, a CPOE down time form to be completed and as per the policy and sent to PathWest laboratory

1. **Samples not suitable for testing:** i.e., haemolysed, clotted, insufficient, unlabelled, mislabelled, unsigned, will not be accepted for testing. The collector will be notified.
2. **Errors in patient identification or labelling:** If you realise or identify an error in patient identification or sample or request labelling, post collection of TM pre-transfusion sample, it is critical that you phone blood bank immediately to inform of error. The consequence of incorrect blood group may be catastrophic.
3. **Neonates:** immediately post birth an identification band with “Mother’s UMRN” is placed on the infant.
4. **Mis-identity or Changed Identity:** If transfusion samples have been taken and/or blood requested when patient identity changes, notify Blood Bank immediately of change (includes baby). A new sample and request will be required.

Error Reporting: Samples and/or requests that do not meet the requirements or do not match cannot be processed. The collector / requestor will be notified to organise a re-collection – a new request form will be required. The errors are recorded by the blood bank and reported to the Standard 7 Blood Management Committee. The most serious error, “Wrong Blood in Tube” where the blood in the sample tube is not from the patient identified on the sample and request, are recorded, and investigated using the Clinical Incident Management System. Blood Bank will initiate the Datix CIMS report

7. Consent

This process applies to all medical officers prescribing transfusion of all blood and blood products to be administered at Armadale Kalamunda Group Hospitals (AKG). AKG medical staff to actively consider the appropriateness and need for transfusing blood and blood products.

Prescribing and ordering blood and blood products, and obtaining and documenting informed patient consent to blood transfusion should be in concordance with:

- [WA Health Consent to Treatment Policy](#)
- [EMHS Patient Consent Policy](#)
- [National Safety and Quality Health Service Standards 2nd Ed, Standard 7 – Blood Management Standard](#)

- [Australian and New Zealand Society of Blood Transfusion Guidelines for the Administration of Blood Products](#)
- [National Blood Authority \(NBA\) Patient Blood Management Guidelines. \(Module 1-5\)](#)
- [Guardianship and Administration Act 1990](#)
- [Human Tissue and Transplant Act 1982](#)

7.1 Informed consent

If the need for a blood transfusion is anticipated, explicit consent must be sought after providing relevant information to the patient unless an exception applies for treatment in an emergency (see the Inability to provide consent section). The consent process must be compliant with the requirements of the Consent to Treatment Policy.

The informed consent discussion must be documented on either the Patient Consent to Treatment of Investigation form (AKMR30) or the Consent to Blood Products form (AKMR176.17). The critical information that must be included in the informed consent discussion includes:

- Reason for the proposed blood product transfusion
 - Risks and benefits of the receiving and refusing the proposed blood product
 - Availability and appropriateness of any other blood management strategies
 - An opportunity for the patient and/or their carer to ask question
 - The patient should receive a copy of the Blood Transfusion Consent Brochure.
- Language and cultural considerations
 - Staff should consider using interpreters and other communication strategies, as necessary; see the [Interpreter and Translator Guideline](#) for further information.

- All Jehovah's Witness patients undergoing a surgical procedure **must** complete either a Consent to Blood Products Form 176.17 or a Release of Liability of Blood Products Form AKMR 176.18
- All non-Jehovah's Witness surgical patients need to be assessed for bleeding risk. If the surgeon determines there is a risk, they must fill out a consent form with the patient.
- In an emergency blood can be administered without consent unless the patient has signed a Release of Liability of Blood Products AKMR 176.18 (or equivalent) form.

7.2 Inability to give consent in medical emergency

Where the patient is *incompetent* and thus not able to consent the circumstances that constitute the medical emergency and the patient's inability to consent must be clearly documented in the patient's health care record post the life-threatening event. When the patient becomes competent, they should be informed of the blood/blood product transfusion and the reasons for providing it.

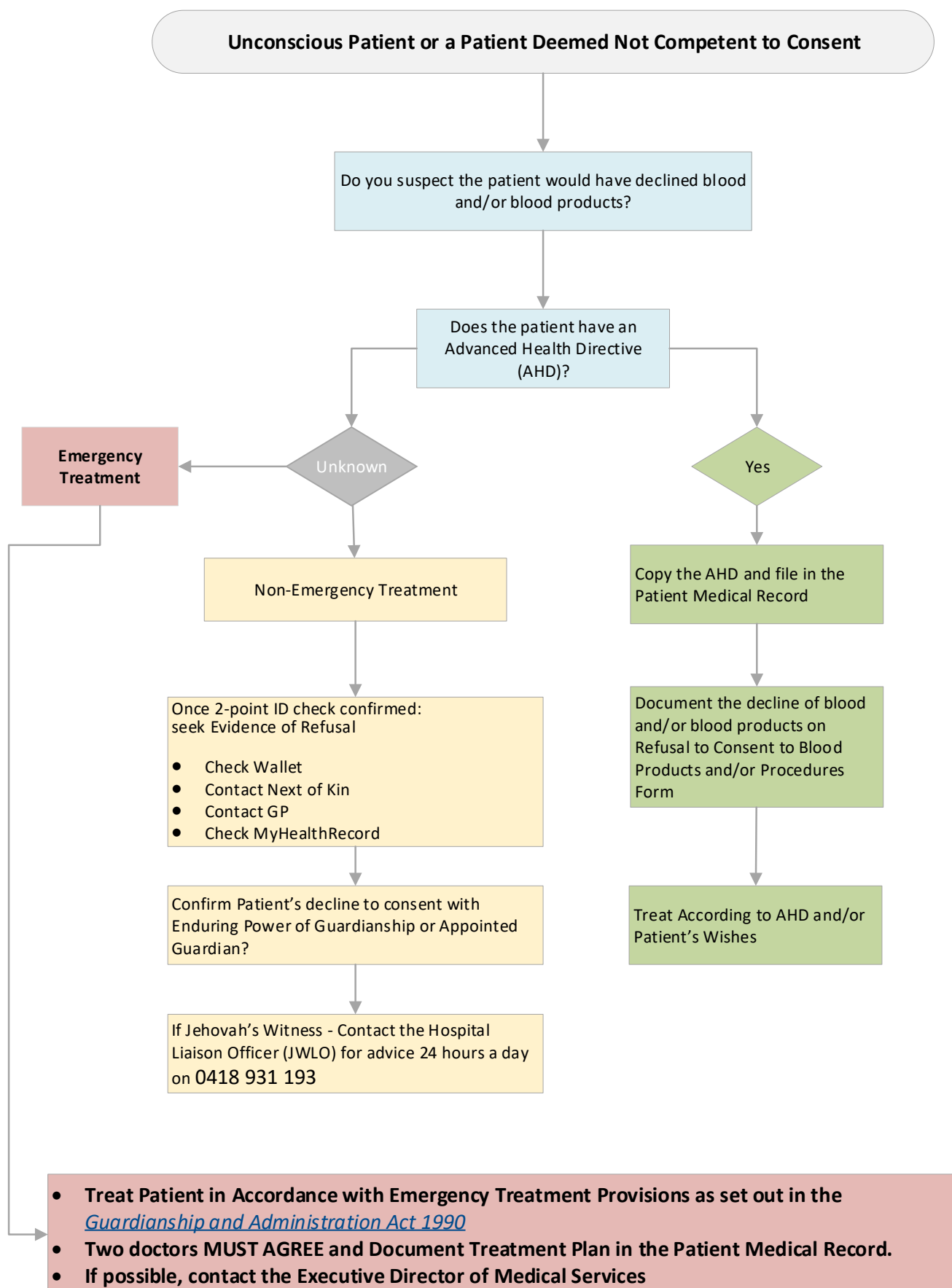
7.3 Consent for an unconscious patient or a deemed not competent to consent who would have declined blood and/or blood products if able

- If an unconscious adult patient is thought to decline blood and/or blood products then an Advance Health Directive (AHD), if previously completed by adult unconscious patient,

should be sought to guide treatment decisions. Once 2-point ID checks confirmed, medical or nursing staff should check the patient's MyHealthRecord, wallet/ purse for a current AHD. The AHD must comply with the requirements of the [Guardianship and Administration Act 1990](#).

- The AHD should clearly state the patient's view on the use of blood and/or blood products, including which treatment they will consent to and which treatments they will decline to consent to.
- The AHD must be copied and filed into the patient's medical record.
If the unconscious patient is thought to decline blood and/or blood products and an AHD has not been located attempts to confirm the patient's treatment decisions should be sought. Once 2-point ID check confirmed, by contacting the patient's Next of Kin or GP
- If these endeavours prove unsuccessful, attempts should be made to see if a substitute decision maker has been appointed to make treatment decisions for the unconscious patient.
- A substitute decision maker is a person who makes a treatment decision on behalf of a patient who is unable to make a reasonable judgement for themselves. The hierarchy of substitute decision makers is defined by [Guardianship and Administration Act 1990](#) **Section 110 ZJ: Order of priority of persons who may make treatment decision in relation to patient** and **110 ZD: Circumstances in which person responsible may make treatment decision**.
- If the substitute decision maker has not been given the Enduring Power of Guardianship or Guardian appointed by the State Administrative Tribunal (SAT), an application to the SAT should be considered.
- If the urgent treatment is required to save the patient's life, prevent serious damage to the patients' health or prevent significant suffering, and a substitute decision maker is not available, the patient must be treated as per clinical necessity in concordance with "Emergency Treatment Provisions" of the [Guardianship and Administration Act 1990](#) **Part 9D section 110ZH: Terms used** and **110ZI: urgent treatment generally** and **110ZIA: Urgent treatment after attempted suicide**.
- All treatment decisions where blood and/or blood products have been administered without consent, or where refusal of consent to blood and/or blood products has been directed by an AHD, the Next of Kin, the GP, the Enduring Guardian or a substitute decision maker for an unconscious patient or in a patient who has been deemed not competent to consent, must be documented in the patient's medical record and on *Refusal to Consent to Blood Products and/or Procedures Form* in conjunction with a treatment plan that has been agreed to by two doctors.

7.3.1 Consent flow chart:



7.4 Refusal: decline to consent to blood and/or blood product

Competent adults have the legal right to decline medical treatment, including the transfusion of blood and blood products. Declining to consent may be based on personal beliefs. But not limited to, Jehovah's Witnesses who have specific beliefs regarding transfusion. However, their acceptance of products may vary according to their personal conscience.

There are several fundamental principles that medical and nursing staff should consider when caring for patients who decline to consent to blood and/or blood products. Medical and nursing staff have an obligation:

- to provide the patient with all the information necessary to enable the patient to make an informed decision and to answer any relevant questions the patient may have
- to satisfy themselves that a patient is fully informed before the patient accepts or declines treatment
- irrespective of their own personal beliefs and opinions, must accept and respect the patient's informed refusal. This includes the patient's right to consent and/or decline specific products, such as consenting to the receipt of fresh frozen plasma whilst declining the use of red cell transfusion.
- must observe specific legal requirements in cases where blood transfusion is indicated for children (aged less than 18 years) during emergency situations. In all cases involving a child patient, the interests and welfare of the child, is paramount and where necessary
- seek legal advice tailored to the specific circumstances of the case

For access to advice for clinicians treating Jehovah's Witnesses a local representative of the Hospital Liaison Committee Network can be contacted 24 hours a day on 0418 931 193

A competent adult patient should provide written documentation of their decline to consent to blood and/or blood products for their Medical Record. Ideally in the form of a current Advance Health Directive.

- An [Advance Health Directive](#) (template available via link) is a legal document in which a person with full legal capacity makes decisions about their future treatment.
- Patients scheduled for elective procedures may have a current AHD form completed, signed, and dated before admission to hospital for their procedure. An AHD is a voluntary document.

Clinical teams should discuss with the patient if there is an [Advance Health Directive](#) available outlining the refusal of consent to the use of blood and/or blood products. As part of the shared decision-making process for care planning the treating medical officer should discuss the availability of non-blood derived medical alternatives and determine an agreed treatment plan with the patient and a Haematologist

7.5 Blood transfusion upon Children without parental consent

A medical practitioner may perform a blood transfusion upon a child without the consent of any person who is legally entitled to authorise the blood transfusion if –

7.5.1 Such person

- fails or refuses to so authorise the blood transfusion when requested to do so; or

- cannot be found after such search and enquiry as is reasonably practicable in the circumstances of the case; **and**

7.5.2 The medical practitioner and another medical practitioner agree

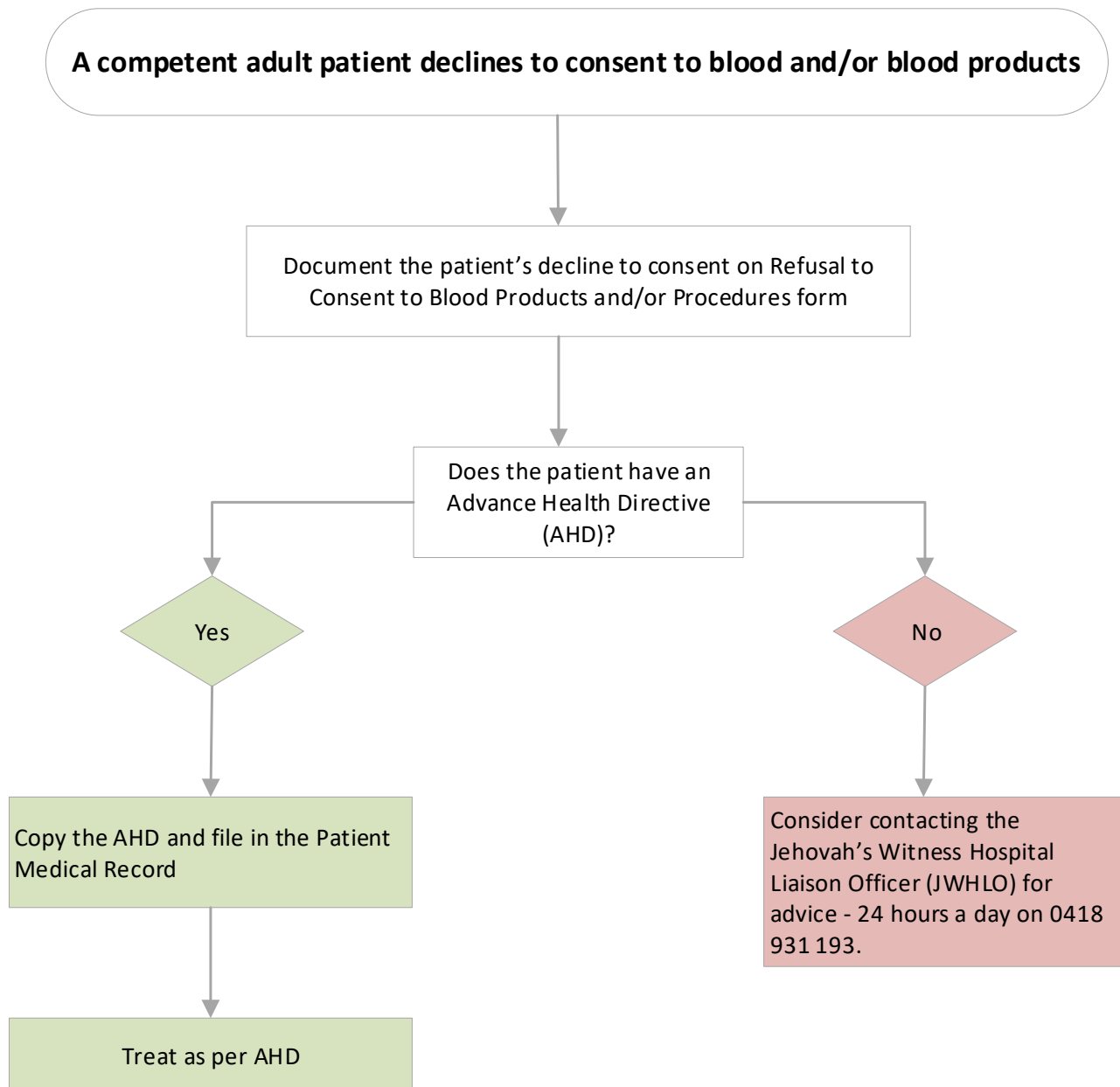
- as to the condition from which the child is suffering; and
- that the blood transfusion is a reasonable and proper treatment for that condition; and
- that without a blood transfusion the child is likely to die; **and**

7.5.3 The medical practitioner who performs the blood transfusion on the child

- has had previous experience in performing blood transfusions; and
- has, before commencing the transfusion, assured himself that the blood to be transfused is suitable for the child.

[Human Tissue and Transplant Act1982 \(legislation.wa.gov.au\)](http://legislation.wa.gov.au)

7.5.4 Declined consent flow chart:



Consult Haematology for red cell optimisation for procedures with anticipated high blood loss

7.6 Validity of consent

The transfusion of blood and blood products can involve more than one course. It must be clearly documented in the patient's healthcare record or on a Consent to Blood Products Form the duration of consent. This may be valid for up to 12 months if the patient does not withdraw consent or there is not a change in the patient's condition.

7.7 Consent Information for Consumers

An information leaflet "Blood Transfusion consent information" is attached to the AKMR 176.17 and can be removed and given to the patient at the time of consent.

Patient consent information for Aboriginal and TSI Patients is available on the Australian Red Cross Lifeblood website page titled "Informed consent" and by scrolling down the page to [Yarning about Blood](#). It can then be downloaded and printed as required.

Resources for non-English-speaking patients are available on the Australian Red Cross Lifeblood [Receiving a transfusion | Lifeblood](#)

8. Prescribing and ordering blood

Consider appropriateness of transfusion and alternatives.

- Discuss plan for care relating to blood and blood products with the patient.
- In red cell transfusion in medically stable patients, consider a single unit to relieve signs and symptoms of anaemia, then reassess.
- Obtain informed patient consent to blood, document on Consent to Blood Products Form - AKMR 176.17
- Determine patient transfusion history (including any previous adverse reactions) and any special requirements i.e. blood warmer, irradiated red cells.
- Ensure current Group and Screen sample available or collect new sample.
- Determine time for transfusion to proceed, based on resources- After hours /Night transfusion should be avoided in medically stable patients. Preferred times for routine and planned transfusions 0800 – 1800.
- Document in patient Medical Record: the proposed blood or blood products, indication, any special requirements, any pre-medications.
- Prescribe on Blood Product Transfusion Form AKMR176.16: include dose, clinical code, and rate of transfusion
- Order blood from PathWest – ensure documentation of any special requirements
- Clearly communicate your request to patient's nurse / medical team
- Post transfusion: reassess patient, document outcome, other investigations and/or treatment required.

8.1 Prescription of blood and blood products

Blood and blood products listed on the [National Blood Authority \(NBA\) National product list](#) are the ONLY products to be prescribed and administered to patients receiving treatment at AKG.

The prescription and administration of Fresh Unrefrigerated Whole Blood (FUWB), also referred to as "Ultra-fresh blood" is **PROHIBITED**, the use of this product does not align with Australian best practice.

A comprehensive definition for FUWB can be found here: [Patient Blood Management Guidelines: Module 1.](#)

The prescriber is responsible for ensuring:

- The blood product transfusion is clinically appropriate
- The expected benefits outweigh the potential hazards
- Informed patient consent has been obtained and documented
- Clinical staff caring for the patient are informed that the blood product has been prescribed
- Patient risk factors are identified, and special requirements are documented
- The prescription is clear and legible

Prescription for blood products must be written on the Blood and Blood Products Form AKMR 176.16 Prescription must contain:

- Patient identification details: Family name, given name, gender, date of birth, UMRN
- Date, Blood Product, Clinical code for Indication to Transfuse, Dose, Rate/Duration, Signature
- Record route of administration i.e., IV or IM etc. where relevant

Post transfusion: record outcome and any other investigations required in medical record.

8.2 Patient special requirements and pre-medications

Document in patient medical record:

- Special requirements e.g., irradiated blood - ensure you inform Blood Bank at time of blood order
- Medications required prior to transfusion
- Special equipment e.g., blood warmer
- Patient transfusion history particularly known previous reactions, or allergies that may be relevant

After-hours transfusions should be avoided between 22:00-08:00 and weekends due to decreased staff numbers in clinical and laboratory areas. The following circumstances may be appropriate for after-hours transfusion:

- Active blood loss
- Symptoms of anaemic compromise e.g., but not limited to shortness of breath, chest pain, tachycardia
- Acute increase in cellular oxygen delivery demand, such as acute coronary syndromes and acute stroke

If after hours transfusion is prescribed, document reason for urgency in patient medical record.

- Acute increase in cellular oxygen delivery demand, such as acute coronary syndromes and acute stroke

8.3 Ordering blood products

Blood may be ordered at time of pre-transfusion crossmatch test request by

- e-order

- CPOE
- downtime yellow pathology form

After initial request, if further blood products are required, please supply a downtime yellow pathology form/with product/s needed and number required.

For collection of the blood product from PathWest the AKMR176.16 Blood Product Transfusion Form will need to be completed and include the following information:

- Full patient identification
- Blood Product, reason for transfusion or the planned procedure
- Date/time required
- Location required
- Your identity
- Number of units

Some blood products may require approval from Haematologist at Fiona Stanley Hospital or at Australian Red Cross LifeBlood (ARCLB) before release e.g., Intravenous immunoglobulin.

Clinical Codes: Indications for Transfusions		
Blood product	Clinical Code	Indication for transfusion
Major Bleed	1	Major Haemorrhage (haemorrhagic shock) inclusive RBC, FFP, PLT, Cryo, Albumin, Prothrombinex, Fibrinogen concentrate, rFVIIa.
Red Cells (RBC)	2	Active Bleeding
	3	Symptomatic anaemia: <i>specify type in medical record</i>
	4	Bone marrow suppression
Platelets (PLT)	5	Bone marrow failure/thrombocytopenia: - Plt count <10x10 ⁹ /L in absence of risk factors
	6	Bone marrow failure/thrombocytopenia: - Plt count <15x10 ⁹ /L in presence of risk factors: <i>specify i.e. fever</i>
	7	Thrombocytopenia: - Plt count <50x10 ⁹ /L with invasive procedure planned
	8	Surgical / invasive procedure: maintain Plt count >50x10 ⁹ /L
	9	Platelet dysfunction related to active antiplatelet therapy or disease state: <i>specify reason in medical record</i>
Fresh Frozen Plasma (FFP)	10	Liver disease in presence of bleeding or at risk of serious bleeding
	11	Multiple coagulation deficiencies: use specific coagulation factors when available
	12	Plasma exchange procedure: <i>specify reason in medical record</i>

Clinical Codes: Indications for Transfusions		
Blood product	Clinical Code	Indication for transfusion
	13	Warfarin reversal in clinically significant bleeding: in addition to Prothrombinex
Cryoprecipitate (Cryo)	14	Disseminated intravascular coagulation
	15	Fibrinogen Deficiency: specify type in medical record
	16	Coagulation factor deficiencies: specify type in medical record
Fractions from Plasma	17	Albumin, IVIG, Prothrombinex, Factors, Immunoglobulins: <i>specify reason in medical record</i>
Recombinant Factors[r]	18	rFVIII, rFIX, rFVIIa: <i>specify factor, product name, dose on prescription and reason in medical record, clearly document MO name</i>

9. Benefits of transfusion may include

- Improved oxygen delivery
- Correct haemostatic defects
- Other benefits depending on blood product prescribed

10. Risks of transfusion may include:

- Transfusion transmitted infections (TTI). All blood donations are collected and tested by the ARCLB for HIV, HCV, HBV, HTLV, Syphilis. ARCLB Blood Component Information 2015 cited risk of TTI from blood as less than 1 in 1 million units. See ARCLB website: [Risks of transfusion-transmissible infection calculated on Lifeblood data](#)
- Adverse reactions. The risk of serious adverse reaction is small. There is no current national data collection system in Australia, but likely incidence based on information from other developed countries is available from ARCLB website - http://www.transfusion.com.au/adverse_transfusion_reactions
- Transfusion related immune modulation, iron overload and other risks.
- Patients requiring further information in regard to blood transfusion may be referred to ARCLB website: www.mytransfusion.com.au

11. Collection, transport, and storage

11.1 Collection of blood products for transfusion

Before blood products are collected, both the patient and the clinician must be adequately prepared to start the transfusion process without delay. Requirements for administration of blood products is outlined in the Administration of Blood Products Procedure.

Where a patient is haemodynamically stable, only one unit of red cells should be removed at a time from a temperature-controlled storage refrigerator, both to avoid wastage and reduce risk. There is an increased risk of wrong transfusion when multiple units for more than one patient are collected. Blood and blood products may be requested/available for collection and transport to the clinical area by designated staff. Product and patient details must be documented on the Blood and Blood Products Record (PathWest)

11.1.1 Before arranging collection of bloods

No consent No hang

- Confirm blood has been prescribed on the *Intravenous Fluid Therapy and Consent*
- Confirm valid consent has been gained, signed by Patient/Guardian and Declaration by Doctor
- Check the patient medical record for transfusion instructions such as pre-transfusion medication or special requirements – see the [PathWest Laboratory Medicine WA Samples and Request Forms for Pre-Transfusion Testing Procedure \(TMP-501\)](#) for further details.
- Ensure that the IV access is patent
- Ensure there is 2 clinical staff available to check blood/blood product and to proceed with transfusion immediately on receipt of product to ward/area
- Carefully check patient identity (URMN, Full name and DOB) against prescription prior to collection.
- Confirm patient prescription (AKMR176) has patient 3 core identifiers and the product required for administration.
- Ensure staff are aware they will need to provide their HE numbers to PathWest on collection of blood or blood product.

11.1.2 Transport of blood products

It is highly recommended that all Patient Care Assistants, Orderlies, or any staff are involved in transporting blood from PathWest to clinical areas complete the e-Learning “Transporting Blood” package available on the AKG My Learning site.

Clinical area/department provides collector with prescription “Blood Products Transfusion Form AKMR 176.16 that has patient 3 core identifiers (i.e., UMRN, full name and DOB), blood product required and patient location – blood cannot be released without this information.

Go to the PathWest Laboratory for issue and release of product

During transport:

- Do not open cooler bags/storage devices used for transport
- Take blood product immediately to the requesting clinical area
- Do not stop for food, drinks, or in public areas, or mix the blood with pharmaceutical products
- Do not place the blood near hot or cold surfaces; do not shake blood
- On arrival in clinical area, hand the blood directly to a nurse/doctor – do not leave blood unattended.
- **Return of blood that is unable to be transfused immediately – the 30 Minute Rule**
- If blood arrives on ward and is unable to be transfused within 30 minutes of dispatch from the PathWest Laboratory, it must be returned to the PathWest Laboratory for storage until the patient is ready.
- Phone PathWest Laboratory and inform staff member of the situation and intention to return blood.

Ensure blood is placed in a clean, opaque bag or cooler bag if available, for return to the PathWest Laboratory – transfer promptly and with same care as delivery



11.1.3 Blood delivered in shipper bags: Only applies to Kalamunda

- PathWest Red Shipper bags are validated for 24 hours and have a temperature logger inside that can be checked that the products inside are within acceptable range.
- Once removed from shipper bag, the infusion must be commenced within 30 minutes. Ensure shipper bag zipper is re-closed post each removal.

[See appendix 2: Shipper bag](#)

12. Management of blood administration

The Blood Book has been developed by Australian Red Cross Lifeblood to assist Australian health professionals in safe bedside transfusion practice

[Australian Blood Administration Handbook](#)

No consent No hang

It is highly recommended that all clinical staff complete the e-learning Clinical Transfusion Practice available on AKG My Learning

It is recommended that all non-emergency blood transfusions are completed 0800 to 1800

Prior to transfusion commencement:

Ensure prescription complete, informed consent documented, indication and special requirements documented, procedure explained to the patient including common symptoms of a possible transfusion reaction.

Ensure IV access is patent; 18-20G is recommended for non-emergency adults. Ensure circumstances/situation appropriate to proceed, including staff availability; avoid transfusing after-hours in medically stable patients

12.1 Blood administration sets

- Blood components must be transfused using an administration set with a filter pore size 170-200 micron.

- Platelets must be transfused through a new blood administration set unless administered in the setting of a massive/rapid transfusion when platelets and plasma may need to be given through the same administration set.
- Platelets must not be transfused through a blood administration set which has been used for red cells as red cell debris may trap infused platelets that may not be ABO compatible.
- Red cells may follow platelets through the same giving set, but not precede platelets
- Blood and other solutions can be infused through the separate lumens of multi-lumen CVCs because rapid dilution occurs in the bloodstream. Where possible, one lumen should be reserved for the administration of blood components.
- Albumin and IV immunoglobulin formulations that do not require reconstitution may be administered via either a standard administration set without a filter, or a blood administration set.

12.1.1 Priming and connecting blood administration sets

- Blood product should be mixed thoroughly by gentle inversion before use
- The blood administration set may be primed with 0.9% sodium chloride or the blood product.
- Blood administration sets must be connected directly to a dedicated IV cannula.
- Attachment to extension tubing (minimum flow 18G) on an IV cannula is acceptable

12.1.2 Flushing blood administration sets

At completion of the transfusion episode, the blood administration set may be flushed with the minimum volume of 0.9% sodium chloride to ensure the patient receives the entire blood product and to maintain IV access if the next red cell unit is not readily available.

For fluids compatible with plasma-derived and recombinant products refer to the individual product information

12.1.3 Changing blood administration sets

The blood component administration set must be changed when the transfusion is completed or every 8hrs if the transfusion episode is not completed.

The blood component administration set must be changed after transfusing 2 units of red cells, during a critical bleed/ massive transfusion this may be exceeded provided the flow rate remains adequate.

12.2 Infusion Methods/devices

12.2.1 Gravity

Gravity flow IV blood administration sets are the preferred method for blood and blood product administration unless otherwise noted in product specific guidelines

12.2.2 Volumetric infusion pumps

Volumetric infusion pumps may be used for administration of blood components. If the use of volumetric infusion pump is required from clinical assessment and/or clinical judgement of the patient (e.g., free flow gravity is unreliable or where controlled rates are required) or local ward preference; the following must be met:

- Blood components which can be transfused via a pump include red cells, platelets, fresh frozen plasma, and cryoprecipitate
- Confirmation that the pump is validated by the manufacturer for blood component administration
- A blood administration set incorporating a 170-200-micron filter is used
- The checking procedure before spiking and hanging the blood must include a check of the device and device settings, as well as the blood product and the patient's identity checks
- The pump settings and volume delivered must be monitored hourly throughout the infusion to ensure that the expected volume is delivered
- Any adverse outcome as a result of using a pump to transfuse blood must be reported as per the Clinical Incident Management Procedure and to the laboratory.
- Volumetric infusion pumps must undergo a regular maintenance program, for example, by the health service biomedical provider.

12.2.3 External pressure devices(bags) and rapid infusion devices:

External pressure devices (bags) and rapid infusion devices are used to assist infusion of large volumes of red cells in the setting of a critical bleeding; rapid infusion devices are also able to warm the red cells. In critical bleeding a large gauge PIVC or CVAD must be used.

External pressure bags, in the absence of critical bleeding, to assist controlled infusion by gravity is not recommended.

External pressure devices (bags) when used should:

- Exert pressure evenly over the entire bag
- Have a gauge to measure pressure
- Not exceed 300mmHg of pressure
- Be always monitored when in use.

12.2.4 Blood warmers

The use of blood warmers may be appropriate in specific clinical situations. The device must be validated by the manufacturer for the administration of blood products and used as specified by the manufacturer.

A blood warmer is indicated for:

- Large volume rapid transfusions (i.e. >50 mL/kg/hour in adults or >15 mL/kg/hour in children)
- Exchange transfusions
- Plasma exchange for therapeutic apheresis in adults
- Patients with clinically significant cold agglutinins
- Trauma situations in which core-rewarming measures are indicated.

General recommendations for blood warmers

- Red cells should only be warmed as they flow through a blood administration set using a specifically designed, approved commercial device with a visible thermometer and audible warning alarm.
- Blood warmers must undergo a regular maintenance program by the health service biomedical provider.

- The operating temperature of the commercial blood warmer must be recorded on the patient's observation record when used to warm red cells.
- Red cells must not be warmed above the set point temperature of the approved device.
- Blood administration sets (incorporating a 170-200-micron filter) used with blood warmers must be primed as for other blood administrations sets prior to use.

Only dry heat blood warming devices should be used due to the risk of contamination from infected water baths.

Alert – Improvised warming techniques:

Improvised warming such as putting the pack in hot water, in a microwave oven or on a radiator must **NEVER** be used. These methods may damage the red cells and cause harm to the patient.

13. Compatible Fluids and Medications:

A multi-lumen access device (i.e., CVC/PICC) is generally safe for continuous co-administration of other therapeutic solutions, allowing for rapid dilution in the blood stream. In the absence of a multi-lumen access device, a second IVC must be inserted if an intermittent or continuous IV infusion is required.

13.1 Incompatible/compatible fluids/medications

Compatible:

- The only IV fluid universally compatible with blood components is 0.9% sodium chloride.
- Red cells are compatible with ABO-compatible plasma and 4% albumin.
- For fluids compatible with plasma-derived and recombinant products refer to the individual product information
- The current formulation of GELOFUSINE® (available in Australia) as stated in the product information

Incompatible:

- Electrolyte and colloid solutions containing any calcium (e.g., Haemaccel®, Hartmann's solution, lactated Ringer's solution) should not be administered with blood components where the anticoagulant contains citrate, as they may cause clotting of the infusion line.
- 5% dextrose in water or hypotonic sodium chloride solutions may cause red cells to haemolyse.

Medications:

- Medications must not be added to the blood bag or the blood administration set / IV line prior to, or during the transfusion.
- Medication may interact with the anticoagulant, additive solutions, or the blood component contained in the bag. A break in the integrity of the infusion line may increase the risk of bacterial contamination of the component. Finally, if an adverse reaction occurs, it is difficult to ascertain whether the medication or the blood component was responsible for the adverse effect.

14. Pre- administration identity check of the patient and blood product

No consent No hang

The final check at the patient's bedside is a vital step in preventing transfusion error. Staff must be vigilant in the checking procedure to ensure that the RIGHT BLOOD is administered to the RIGHT PATIENT

14.1 Patient Identification

All patients receiving a blood product must be positively identified and should have a secure identification band attached to their body in accordance with the EMHS Patient Identification Policy.

The following core patient identifiers are mandatory:

- Family Name and Given Name(s)
- Unit Medical Record Number (UMRN)
- Date of Birth (written as DD/MM/YYYY)

14.2 Unconfirmed identity and name changes

- Where a patient's identity cannot be reliably confirmed they must be registered as 'Unknown Male' or 'Unknown Female' using an emergency UMRN generated by WebPAS.
- Once the patient has been identified, patient information is to be updated in WebPAS and a new identification band is to be attached to the patient.
- If a patient is incorrectly identified and the identification band issued contains incorrect patient details, it is to be removed and patient information is to be updated in WebPAS and a new identification band is to be attached to the patient.
- All departments that have performed investigations and treatment, such as imaging, pathology, and pharmacy, must be notified as a matter of urgency.
- If the PathWest Laboratory has already performed tests (Group and Screen or cross matched blood) they must be notified immediately. See Section 3, Section 11, and Appendix A of the EMHS Patient Identification Policy.
- A written entry must be made in the patient health record of the change in identity and the time it occurred.

Alert – Patient identification and critical bleeding:

- If the identity is confirmed during a critical bleeding situation or critical medical management, e.g., resuscitation or surgery, then the change of identity should occur only when the period of potential/actual critical bleeding has ended and no longer requires transfusion support.

14.3 Checking procedure

- The patient's identity must always be confirmed prior to transfusion with the patient's full name (family and given name), UMRN and DOB, required to be checked and found to be identical on all the following:
 - Patient's identification band
 - The compatibility label attached to the blood product

- The blood product prescription.
- Two (2) staff members (competent in the administration of Blood and Blood Products) must undertake the identity check of the patient and the blood product, independently, at the patient's side immediately prior to administration. Each of these two staff are responsible for the accuracy of the checking procedure.

The staff performing the check must be medical officers, registered nurses or midwives, anaesthetic technicians or other staff authorised and appropriately trained by their health service in accordance with the AKG Medication Administration policy.

- The two individuals carrying out the check must both sign the Blood Product Transfusion Form confirming the patient and product check has occurred and is correct and compatible
- The person spiking/hanging the blood product must be authorised to spike/hang the product and should have completed the BloodSafe eLearning: Clinical Transfusion Practice. They must be one of the two staff members who have undertaken the blood and patient identity check. The pack should not be spiked until the identity check of the patient and product is complete. The pack must be spiked and commenced immediately after the check has been completed. If there is a delay, the checking process must be repeated.
- It is the responsibility of the person spiking/hanging the blood product to ensure it is appropriate to undertake the transfusion at that time e.g., patient's clinical status, sufficient staffing, and/or confirmation with the prescriber.
- The secondary compatibility label must be removed from the blood product when the pack has been commenced and placed in the designated area on the Blood Product Transfusion Form confirming the product has been transfused and enabling traceability if required.

14.4 Blood product checklist

The blood product checking procedure must ensure the following:

- The patient's full name (family and given name), DOB, and UMRN are identical to those on the compatibility label attached to the blood product and the blood prescription.
- The blood product type is the same on the prescription, on the product and the laboratory compatibility label.
- The product is checked for compliance with any special requirements on the prescription (e.g., irradiated, CMV seronegative).
- The blood group and donation/batch number on the compatibility label are identical to that information on the product from the supplier.
- The blood group on the blood component is compatible with the blood group of the patient as indicated on the compatibility label attached to the pack; if the blood group of the blood component and the patient are not identical, the transfusion service provider must make a specific comment to indicate that it is compatible (or most suitable available).
- The blood component/product has not passed its crossmatch expiry or unit/product expiry date and time by the commencement of the transfusion to the patient.
 - The integrity of the blood product is confirmed by excluding:
 - Any leaks at the ports and pack seams
 - Any leaks at the ports and pack seams
 - Any evidence of haemolysis, unusual discolouration, or turbidity
 - The presence of any large visible clots

- Broken or leaking bottles or vials.

If a discrepancy is found while performing the checking procedure at the patient's side, that is not covered by a comment by the issuing transfusion service provider, the blood product must not be transfused until the discrepancy is resolved with the transfusion service provider.

If there is any concern regarding the integrity of the product, it must not be used and should be returned to the issuing transfusion provider.

Blood checking procedure

- ✓ Patient **MUST** have Identification (ID) band attached
- ✓ Informed blood consent **MUST** be obtained
- ✓ Two (2) staff members **MUST** carry out ID check of patient & blood **INDEPENDENTLY** at the **BEDSIDE** immediately prior to administration and one (1) must be an RN or Medical Officer
- ✓ Person spiking/hanging blood pack **MUST** be one (1) of the two (2) staff members performing the check.

1. Patient details
a) Ask patient to state: Family Name, Given Name in full & DOB
b) Patient ID band details
c) Details on back of blood pack: Family Name, Given Name in full, DOB & UMRN
ALL patient details MUST be identical

2. Donation number on front & back of pack must be identical

3. Component description must be consistent with medical prescription

4. Blood group on front and back of pack must be compatible

5. Expiry date on pack label

6. Check bag for leaks, discolouration or clots

BACK: Compatibility Label

FRONT: Blood Pack Label

If there is any discrepancy, **DO NOT TRANSFUSE**. Phone Pathwest extension 2031

14.5 Infusion rates and observations

The infusion rate for blood products depends on the clinical context, age and cardiac status of the patient. In stable, non-bleeding adult patient's typical administration durations are:

Blood product	Commencement of transfusion	Transfusion rate	Compatibility	
Packed Red Cells	Within 30 minutes of collecting from Blood Bank	Usually over 60-180 minutes. Each unit must be completed within 4 hours	Should be Rh(D) matched whenever possible but Rh(D) positive patient can have Rh(D) negative red cells.	
			Patient Group	Compatible Group
			A	A, O
			B	B, O

Blood product	Commencement of transfusion	Transfusion rate	Compatibility	
			AB	AB, A, B, O
			O	O
Platelets	Usually given stat to 30 minutes	15-30 minutes per standard adult equivalent dose Each unit must be completed within 4 hours	ABO identical platelets preferred. Non- identical ABO platelets may be provided. Phone your transfusion service to confirm.	
Fresh Frozen Plasma	Usually given stat to 30 minutes	30 minutes per unit (i.e. 10-20mL/kg/hr). Each unit must be completed within 4 hours	No need to Rh(D) match.	
Cryoprecipitate	Usually given stat to 30 minutes	30-60 minutes per standard adult dose (i.e. 10-20mL/kg/hr) Each unit must be completed within 4 hours	Patient Group	Compatible Group
			A	A, AB
			B	B, AB
			AB	AB
			O	AB, A, B, O

- Plasma-derived products should be infused in a timeframe in accordance with product-specific instructions.
- For patients at risk of circulatory overload (e.g., cardiac failure) it is usually necessary to transfuse more slowly with frequent monitoring. Concomitant use of diuretics should also be considered.
- Patients with acute/critical bleeding or who are in hypovolemic shock require blood components to be transfused rapidly. The use of a blood warmer is recommended in critical bleeding / massive transfusion situations.
- Commence each blood component pack within 30 minutes of removal from the controlled storage or if there is a delay in commencement then return to the transfusion service provider within the 30-minute timeframe.
- Start each pack slowly, where possible and clinically appropriate. The rate of infusion may then be increased, usually after 15 minutes, to the maximum infusion rate defined in accordance with the prescription if there are no signs or symptoms of an adverse reaction.

14.6 Maximum duration for transfusion

The transfusion should be completed within four (4) hours of the product leaving the approved controlled storage (or sooner if specified on the pack / by the transfusion service provider) and no longer than 4.5 hours following release of the product from temperature-controlled storage.

Note: The following recommendations relate to transfusions of blood components only. While the general principles apply to plasma-derived products refer to relevant product information and specific product guidelines as stated in the Blood Transfusion Manual.

14.7 Monitoring

- Prior to commencement of the transfusion the patient should be educated and advised to report to staff immediately any adverse effects that they may experience during or after the transfusion. Ensure patient has the Blood Transfusion Consent Information which includes advice post transfusion.
- The patient must be monitored for signs of potential complications/adverse reactions, and any suspected problems dealt with promptly.
- The patient must be observed for the first 15 minutes after commencement of each unit and observations should continue throughout the duration of each unit and post completion of the transfusion, to detect any adverse effects, in accordance with the observations section of this procedure. When the patient is not in an open area that allows continual observation consideration should be given to attending the patient for the first 30 minutes.
- Patients receiving a Platelet transfusion, the nurse/midwife must remain with the patient for the duration of the transfusion.

14.8 Observations

- The commencement time of the transfusion must be documented.
- Vital signs must be recorded on the Adult Observation and Response Chart or equivalent appropriate to specific areas (e.g., ICU, ED, Maternity wards). Ensure the Adult Deterioration Detection System (ADDS) score is documented as this will alert the staff to the more serious and life-threatening transfusion reactions.
- Baseline vital signs of temperature, pulse, respiratory rate, and blood pressure must be documented prior to commencement of each individual blood component pack administered.
- Subsequent minimum vital signs must be documented at 15 minutes, hourly and on completion, per individual pack as follows:
 - RBC, FFP and Cryoprecipitate – temperature, pulse, respiratory rate, and blood pressure
 - Platelets – temperature, pulse, respiratory rate.
- The frequency and documentation of vital signs should be adjusted according to the patient's clinical condition based on underlying co-morbidities and current clinical factors e.g., if the patient has congestive cardiac failure, bleeding, increased intracranial pressure, renal dysfunction or is unable to respond or show signs or symptoms of a reaction more frequent monitoring may be appropriate.

14.9 Completing the transfusion

Document the time each product was completed.

- At the completion of a transfusion flush to clear the line and to ensure the patient receives the entire product.
- The compatibility label must be retained in the patient's medical record, Blood Product Transfusion Form.
- Document the outcome of the transfusion in the patient's medical record.
- Adverse effects may manifest after the transfusion has been completed. The patient must be advised to report any adverse effects experienced after the transfusion has been completed.
- For patients undertaking a transfusion in a day treatment ward/area consideration, dependent on the patient's clinical status and transfusion history (if applicable), of a period of continued

observation at the completion of a transfusion may be appropriate, in case of a delayed transfusion adverse event.

- If there is any suspicion of a transfusion reaction, the event must be reported and the blood component pack returned to the transfusion service provider, and the event documented in the patient's medical record as per the [PathWest Laboratories Medicine WA: Transfusion Reaction Investigation Procedure \(TMP-515\)](#).

If the transfusion is completed uneventfully, discard the empty pack as per the [EMHS Waste Management Policy](#) for disposal of clinical waste.

15. Adverse Reactions and Haemovigilance Reporting

Transfusion while beneficial can also present potential risks for a patient. Adverse transfusion reactions are when patients experience an undesirable response associated with transfusion of blood or blood products.

Patients who have previously been transfused, multiparous women and patients receiving emergency uncross matched transfusion are at increased risk of immediate and delayed haemolytic transfusion reactions due to the presence of antibodies.

As soon as you suspect a transfusion reaction:

- Stop the transfusion immediately and activate emergency procedures if required.
- Check and monitor the patient's vital signs.
- Maintain intravenous (IV) access (do not flush the existing line and use a new IV line if required).
- Repeat all clerical and identity checks; ensure the right pack has been given to the right patient.
- Notify your Medical Officer and Transfusion Service Provider.

After the transfusion has been stopped you may be required by PathWest to send freshly collected blood and urine samples along with the component pack and IV line. Always follow occupational health and safety policies (e.g., do not transport IV lines with the insertion spike exposed).

The National Blood Authority (NBA) has developed a reporting and governance framework for the National Haemovigilance Program for Australia. It reports on serious transfusion related adverse events occurring in participating public and private hospitals.

[Haemovigilance Reporting Flowchart](#)

16. Massive transfusion protocol - Obstetric

Obstetric – see [appendix 3](#)

17. Massive transfusion protocol – Non - Obstetric

Non-Obstetric – see [appendix 4](#)

18. Fibrinogen Concentrate

RiaSTAP or Fibrinogen (Factor I) Concentrate

18.1 Description

Concentrated human factor I (fibrinogen)

Fibrinogen concentrate is an alternative to cryoprecipitate to replace fibrinogen.

18.2 Indications

A low fibrinogen level, or an expected low fibrinogen level in an acute setting.

18.3 Administration/Dosing

- Intravenous (IV)
- The dose of fibrinogen concentrate may be empiric (70 mg/kg), based on either the laboratory fibrinogen concentration or thromboelastometry.
- Adults:
 - Generally: 2-4 mg (1gm vials) for average-sized adult
 - If fibrinogen level unknown, administer **70 mg/kg**
 - Dosing suggested by manufacturer:

$$\text{mg/kg} = [(\text{Target fibrinogen level g/L}) - (\text{Actual fibrinogen level g/L})] \div 0.017$$
 E.g., $\text{mg/kg} = [2.2 \text{ g/L} - 1.0 \text{ g/L}] \div 0.017 = 70 \text{ mg/kg} \rightarrow \times 75 \text{ kg} = \sim 5.3 \text{ g}$
- After reconstitution needs to be used within 8 hours.
- Children:
 - Haematology led administration.
 - Suggestion of 30-60 mg/kg and consultation with the relevant haematology service.

18.4 Administration

- Manufacturer recommends 5 mL/min, and other guidelines recommend 20 mL/min (i.e. slow push over 2.5mins)
- Can be administered peripherally as with all other blood products
- **Not compatible with any other blood products nor other medications**
- Flush with saline
- Tips on how to reconstitute are below.
- See manufacturer insert for reconstitution guide

18.5 Group/Crossmatch

- Not required

18.6 Adverse Effects

- Thromboembolic episodes including MI, PE, DVT and arterial thrombosis
- Allergic reaction: anaphylaxis, dyspnoea, rash
- Chills, nausea, vomiting
- Risk of infectious disease transmission such as: CJD, HIV, Hep B and C

18.7 Contradictions

- Known hypersensitivity / anaphylaxis reactions to this product or other concentrated fractionated plasma products

18.8 Advantages and disadvantages of Fibrinogen to Cryoprecipitate

18.8.1 Advantages

- Does not need to be thawed and so can reduce the time it takes to administer fibrinogen replacement
- No grouping / crossmatch required
- Much more convenient to give a large amount of fibrinogen in a relatively small volume
- Useful in pre-hospital or regional settings where blood product storage is limited
- Long shelf-life of 60 months (vs 12 months for cryoprecipitate)

18.8.2 Disadvantages

- Costly, roughly double the cost of
- Lack of familiarity

18.9 Fibrinogen Location

- Pharmacy (Fridge 3) – after hours, contact after hours Clinical Nurse Manager * 4316 to access
- Theatre Automated Drug Fridge
- Intensive Care Automated Fridge

18.10 Reconstitution tips

- You will need, in addition to the package contents, a 50-60 mL luer-lock syringe, and a sterile alcohol wipe (or equivalent)
- You will also need a decent amount of bench space if you are reconstituting more than 1gm

19. Intravenous Iron Transfusion

Iron is an essential element required for the formation of haemoglobin and myoglobin.

Oral iron therapy is suitable and effective as first line therapy in most patients, including most obstetric patients

[National Blood Authority Iron product choice and dose calculation for adults](#)

[AKG Intravenous Iron Therapy- Adult Guideline](#)

[AKG Administration of Intravenous Iron Polymaltose for Haemodialysis Patients Procedure](#)

[AKMR 197 Intravenous Iron Protocol for Haemodialysis Patients](#)

[AKG Iron Infusion Consent and Checklist AKMR30.11](#)

[AKG Consumer Information Intravenous Iron](#)

20. Training Requirements

Armada Kalamunda Group strongly recommend all clinicians involved in the consenting, prescribing, administration and monitoring of blood transfusions complete the Clinical Transfusion Practice Course (by BloodSafe eLearning Australia)

This course is based on current clinical guidelines, evidence-based material and/or expert consensus opinion and have been developed in collaboration with leading transfusion medicine experts in Australia.

For Patient Care Assistants or Porters involved in the Transport of Blood from PathWest it is strongly recommended that they complete the Transporting Blood Course (by BloodSafe eLearning Australia).

Both courses are available on the MyLearning Portal [MyLearning](#)

21. Blood Shortage

The Business Continuity Management sets out the principles to be followed, actions to be taken and the resources to be used when responding to a critical incident. Statewide blood shortage is included in this plan.

- [AHS Business Continuity Plan State-wide blood shortage page 117](#)
- [National Blood Supply Contingency Plan](#)

22. Compliance monitoring

Compliance against this procedure will be monitored by the AKG Blood Management Committee.

- Compliance monitoring activities relevant to this procedure are included in the Terms of Reference for the Blood Management Committee

23. Policy context

The following documents are required to give effect to this procedure:

- Legislative
 - [Guardianship and Administration Act 1990](#)
 - [Human Tissue and Transplant Act 1982](#)
- HDWA
 - [Consent to Treatment Policy](#)
 - [Clinical Incident Management Policy](#)
- EMHS
 - [Patient Identification and Procedure Matching](#)
 - [Patient Consent Policy](#)
 - [Waste Management Policy](#)
- AKG
 - [Intravenous Iron Therapy – Adult Guideline](#)
 - [Administration of Intravenous Iron Polymaltose for Haemodialysis Patients Procedure](#)
- Other
 - [Australia and New Zealand Society of Blood Transfusion Guidelines](#)
 - [Guidelines for the Administration of Blood Products](#)
 - [Guidelines for Transfusion and Immunohaematology Laboratory Practice](#)
 - [Guidelines for the Laboratory Estimation of Fetomaternal Haemorrhage](#)
 - [Guidelines for Extended Life Plasma: A Framework for Preparation, Storage and Use](#)
 - [Australian Standard AS3864.1-2012 Medical Refrigeration Equipment – For the Storage of Blood and Blood Products](#)

- [Blood component information June 2019](#)
- [BloodSafe e-Learning](#)
- [AHS Business Continuity Plan State-wide blood shortage page 117](#)
- [National Blood Supply Contingency Plan](#)
- [Women and Newborn Health Service Transfusion Protocols \(Neonates\)](#)
- [National Blood Authority \(NBA\) - Patient Blood Management Guidelines](#)
- [Iron product choice and dose calculation guide for adults | National Blood Authority](#)
- PathWest Laboratory Medicine WA:
 - [Transfusion Blood Refrigerator and Blood Product Freezer Policy \(TMP-525\)](#)
 - [Blood and Fridge Failure Policy \(TMP-062\).](#)
 - [Blood Product Storage Equipment Failure Policy \(TMP-531\).](#)
 - [Tracking the Fate of Transferred Blood Products Policy \(TMP-539\)](#)
 - [Transportation of Donor Blood Procedure \(TMP-537\)](#)
 - [Receival of Red Cell Products by non-PathWest staff Procedure \(TMP-083\).](#)

24. Supporting information

The following documents support the implementation of this procedure:

- [Module 1 Critical Bleeding/Massive Transfusion](#)
- [Module 2 Perioperative](#)
- [Module 3 Medical](#)
- [Module 4 Critical Care](#)
- [Module 5 Obstetric and Maternity](#)
- [Module 6 Neonatal and Paediatrics](#)

25. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 7

26. References

1. **Human Tissue and Transplant Act 1982** Donations of tissue by living persons Part II Blood transfusions Division 5 s. 21 As at 19 May 2017 Version 03-a0-03 page 13
[Australian Commission on Safety and Quality in HealthCare. National Safety and Quality Health Service Standards, Standard 7 Blood Management, 2021](#)

27. Document owner

Executive Director - AKG

Enquiries relating to this procedure may be directed to:

Title: Director Clinical Services
Department: Executive
Email: Tania.Strickland@health.wa.gov.au

28. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

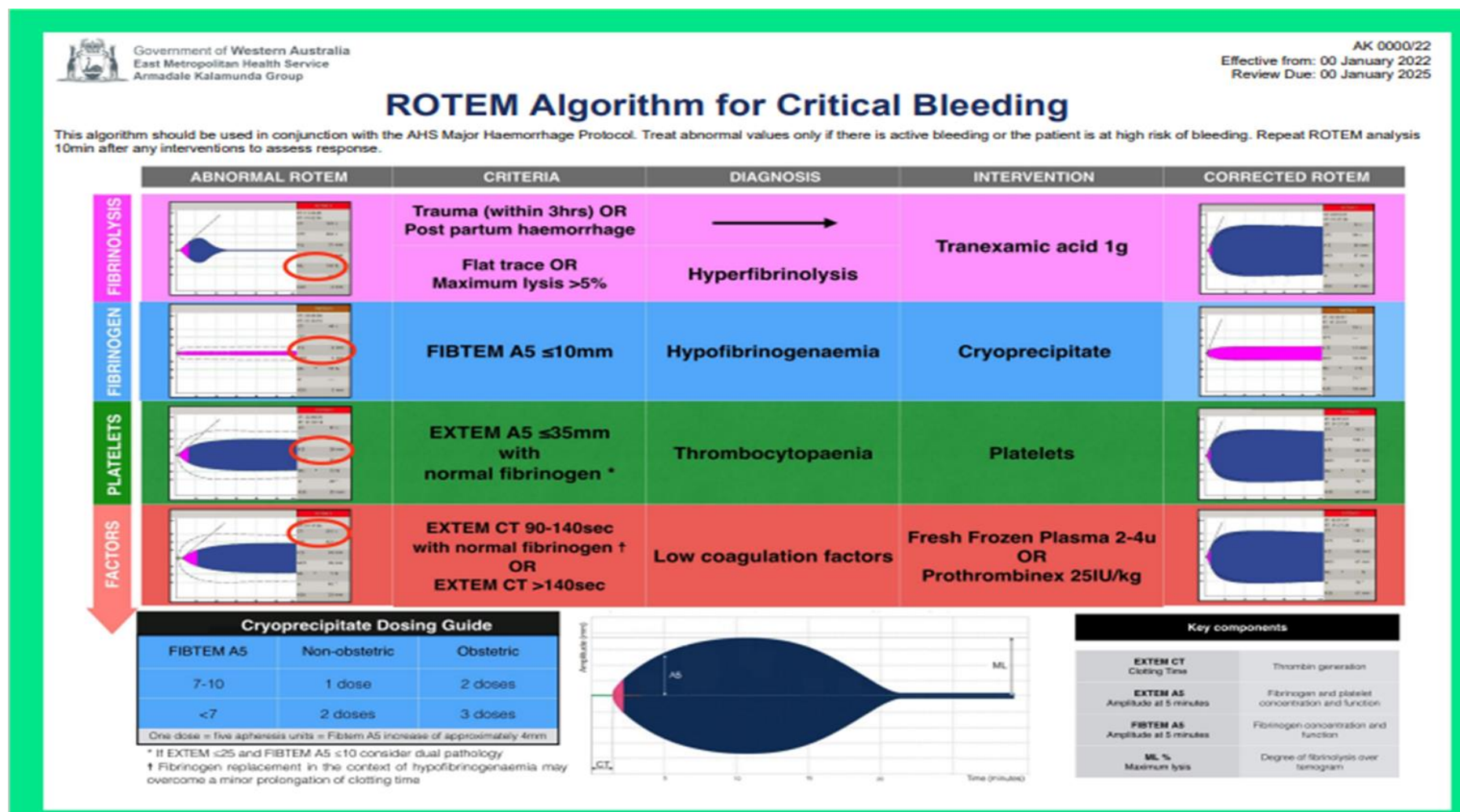
Version	Effective from	Effective to	Amendment(s)
V2.1	23 01 2026	28/08/2027	Minor Review to update Obstetrics and Non-Obstetrics Massive Transfusion Protocol as per Lifeblood

29. Authorisation

This procedure has been authorised and issued by:

Authorised by	Alisha Thompson - Executive Director AKG
Authorisation date	23 01 2026
Published date	23 01 2026

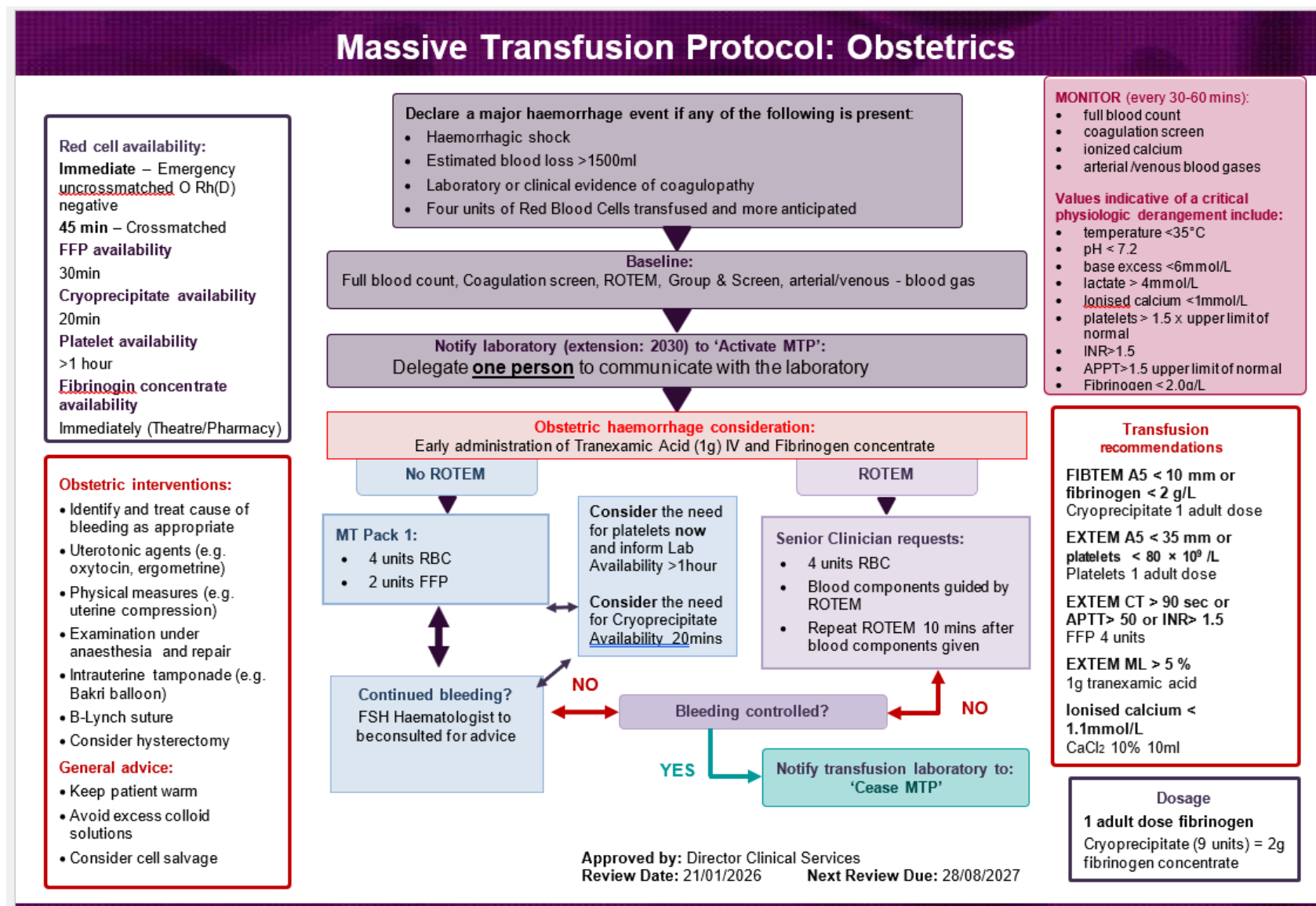
Appendix 1 ROTEM



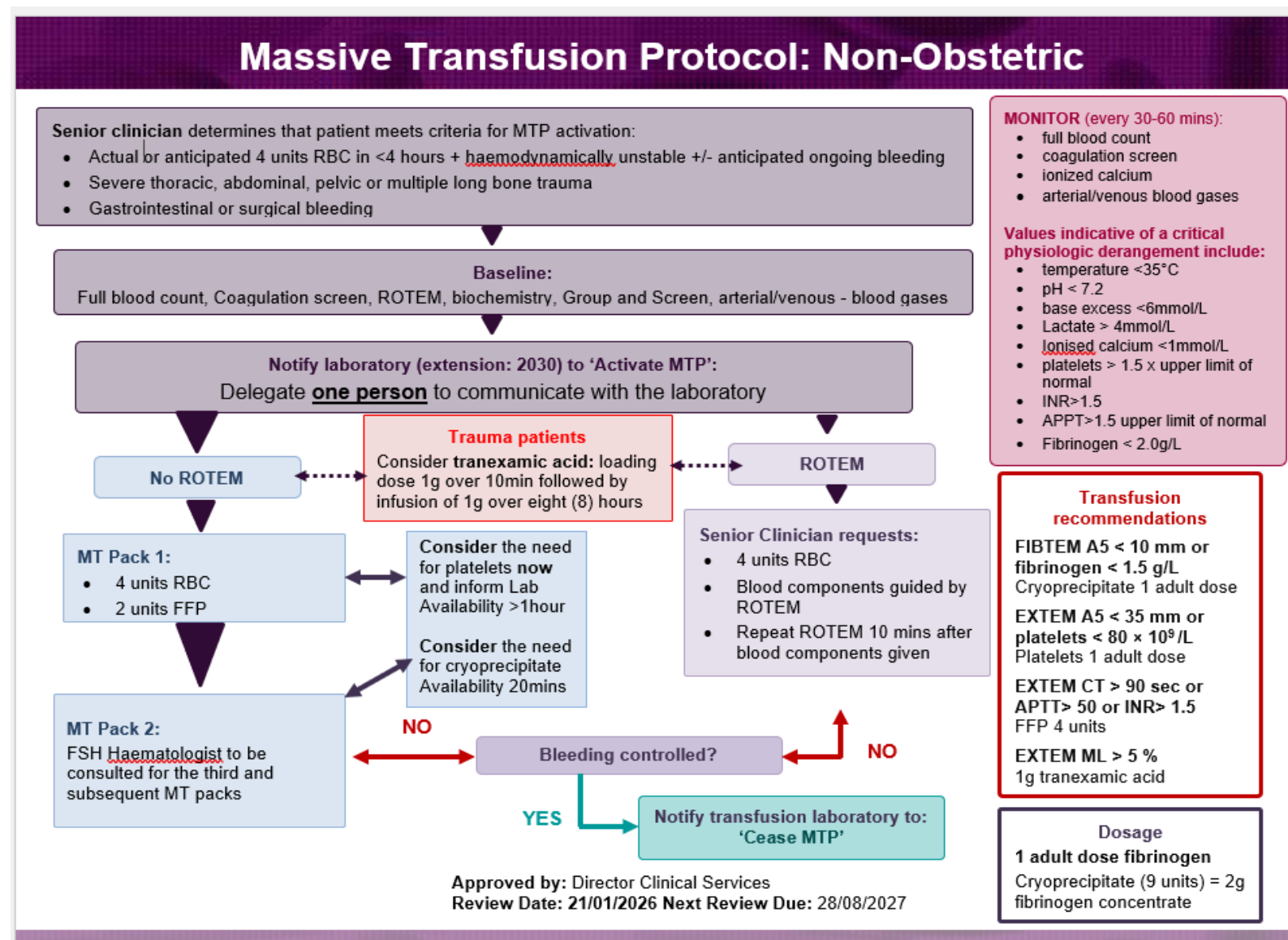
Appendix 2: Shipper Bag



Appendix 3: Massive transfusion protocol: Obstetrics



Appendix 4: Massive transfusion protocol: Non-Obstetric



Appendix 5: Haemovigilance reporting flowchart

Haemovigilance Reporting Flowchart

- Haemovigilance is widely recognised as an integral part of blood transfusion safety. Transfusion-related adverse events can include reactions to administered blood products as well as clinical incidents related to the delivery of health care.
- Refer to the [PathWest Laboratory Medicine WA: Transfusion Reaction Investigation Procedure \(TMP-515\)](#) and the WA Haemovigilance Reporting Guideline for further details.

